

KEMSA Supply Chain Intelligence Platform — Business Case

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Problem

KEMSA distributes medical commodities to public health facilities across all 47 counties. Today, two things happen too late: facilities run out of essential medicines before anyone is warned, and counties fall badly behind on payments without early warning. Redistribution of surplus stock between facilities is still a manual, slow decision — so stock expires in one place while another runs dry.

Proposed Solution

A three-part decision-support platform, already built and tested against real historical data:

- Risk Scorecard — ranks every county's payment risk automatically (e.g. flags a county at 92/100 risk vs. one at 6/100), with a plain-language reason attached.
- Stockout Forecasting — predicts, per facility and medicine, when stock will run out (30/60/90 days ahead) instead of finding out after the shelf is empty.
- Redistribution Matching — automatically finds the nearest facility with surplus stock and proposes a transfer, with a human approving or rejecting before it happens.

Financial Impact (Year 1, conservative)

Demonstrated opportunity (from one quarter of real 2024 data, ~85,000 transfer matches)	KES 49.2M / quarter
Annualized opportunity (same trend, 4 quarters)	KES 196.8M
Year-1 capture assumption (phased rollout, one region first — conservative 10%)	KES 19.7M
Platform cost (Year 1: hosting + part-time maintenance, ~KES 200k/month)	KES 2.4M
Net Year-1 benefit (KES 19.7M – KES 2.4M)	KES 17.3M
Return on Investment (Net benefit ÷ Cost)	~720%
Payback period (Cost ÷ average monthly benefit)	~1.5 months

Note: the KES 49.2M figure is not a projection — it is what the historical data shows was already recoverable in a single quarter. The 10% Year-1 capture rate is a deliberately conservative assumption for a phased, single-region rollout; it does not assume full national adoption in Year 1.

Top Risks & Mitigation

- Adoption risk — county and facility staff may not trust automated recommendations. Mitigation: every redistribution suggestion requires human Approve/Reject before it executes; nothing moves stock on its own in Year 1.
- Forecast accuracy risk — predictions depend on consumption data quality, which varies by facility. Mitigation: forecasts are tiered (Critical/Warning/Watch/Healthy) rather than binary, and every high-value transfer is reviewed before approval.

Recommendation: GO

The underlying value has already been demonstrated in real historical data, not modeled from assumptions. Cost is low relative to the demonstrated opportunity, risk is controlled by keeping a human in the approval loop, and payback is achieved within the first two months of a single-region pilot. Recommend proceeding to a one-county pilot, with a Go/No-Go review on national rollout after 90 days of live data.